

Yash Pathak

✉ yp290304@gmail.com ☎ +91 7045820493 🌐 Github 💻 LeetCode in LinkedIn 📁 Portfolio

Education

St. Francis Institute of Technology, Mumbai

B.Tech in Electronics and Telecommunication Engineering with Honors in AI/ML

Nov 2022 – May 2026

CGPA: 7.8 / 10

Technologies

Languages: Python, Java, C, Dart

AI/ML Frameworks: Tensorflow, PyTorch, OpenCV, Scikit-learn, LangChain, HuggingFace

Version Control and OS: Git, GitHub, Linux

Tools and Platforms: Streamlit, Figma, SQL(BigQuery, MySQL), GCP, AWS, Spring Boot, Twilio, MQTT

Coursework: Data Structures and Algorithms, DBMS, Computer Networks

Projects

GYANTRA [↗](#)

2025-26

- Built an autonomous research agent using LangChain that breaks complex queries into structured multi-step execution plans through a self-reflective ReAct loop with loop detection and step-limit controls to prevent runaway execution.
- Designed a modular tool layer covering web search (Tavily and Exa), Wikipedia retrieval, live URL scraping and SymPy-based math execution supporting 50+ domain-specific expert prompts across academic, business and technical research.
- Tools Used: Python, LangChain, OpenAI API, Anthropic API, Tavily, Exa, SymPy

FLIP-CHECK FLIPKART GRID 6.0 [↗](#)

2024-25

- Developed a YOLO-based freshness detection system using OpenCV that classifies fruits and vegetables with 96% accuracy and integrated barcode scanning, OCR and object counting into a multi-feature real-time dashboard.
- Tools Used: Python, Streamlit, OpenCV, PyZbar, Tesseract, Roboflow SDK

VOICEOPS

2026 - Ongoing

- Building a voice AI customer support platform on Java Spring Boot with fine-tuned in-house STT and TTS models replacing third-party speech APIs and integrating Twilio for inbound and outbound telephony handling across the call lifecycle.
- Fine-tuning a multilingual LLM trained on Indic language data for conversational AI responses and designing a Next.js CRM-style dashboard that surfaces live call data, transcripts and customer interaction history in a unified interface.
- Tools Used: Java, Spring Boot, Next.js, Twilio, PyTorch, REST APIs.

Experience

Software Engineer Intern

1Accord

Mumbai

July 2025 – Present

- Architected an integrated rifle and pistol shooting system with a proprietary Lakshya target machine, a 94% accurate image processing model for automated scoring and a real-time Next.js dashboard connected via low-latency TCP over WiFi.
- Developed a Python-based technical information system for Darkstores implementing MQTT integration to improve data communication reliability.
- Designed a BLE-enabled pet safety application streamlining device-to-device synchronisation for real-time tracking.

Leadership/Extracurricular

Project Developer

Team Robotics And Automation Wing (TEAM RAW)

Mumbai

July 2023 – Sept 2025

- Designed and built AI/ML-driven autonomous robots integrating PID controllers and real-time object detection; achieved 95% detection accuracy using YOLO and improved task execution accuracy by 40%.
- Led AI/ML and control system development in ABU ROBOCON, E-Yantra and ISRO IROC securing AIR 9 in design in ISRO IROC and advancing to semi-finals in E-Yantra.